

ANTOINE SCHNEPF

Criteo AI Lab, Paris · Université Côte d’Azur, CNRS, I3S, Nice
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PROFILE

Ph.D. candidate in Machine Learning at Université Côte d’Azur and Criteo AI Lab, working on **neural 3D reconstruction and generation with 2D visual priors** — the subject of my thesis, to be defended in July 2026. My research combines a rigorous mathematical background (CentraleSupélec, MVA at ENS Paris-Saclay, *classes préparatoires* in Strasbourg) with four years of applied research experience in industry and academia. I have published four peer-reviewed papers, including two at **ICLR**, one at **ICML**, and one at **TMLR**. I bring strong mathematical maturity, solid engineering practice, and a track record of independent contribution to state-of-the-art 3D computer vision.

RESEARCH INTERESTS

- Neural 3D reconstruction — NeRFs, Tri-Planes, latent-space representations.
- Generative 3D modeling — diffusion, GANs, and panorama-based world generation.
- Bridging 2D visual priors with 3D inverse graphics.

EDUCATION

Ph.D. in Machine Learning

2023 – 2026 (expected)

Nice / Paris, France

Université Côte d’Azur & Criteo AI Lab

- *Thesis.* *Neural 3D reconstruction and generation with 2D visual priors.*
- *Advisors.* Andrew Comport (Université Côte d’Azur, CNRS, I3S), Flavian Vasile (Criteo AI Lab).
- *Topics.* Neural radiance fields, generative 3D modeling, 2D computer vision.

M.Res. MVA — Mathématiques, Vision, Apprentissage

2021 – 2022

Paris, France

École Normale Supérieure Paris-Saclay

- *Coursework.* Deep Learning, Reinforcement Learning, Statistical Learning, Topological Data Analysis, NLP.

M.Sc. Engineering, Major in Mathematics

2018 – 2022

Paris, France

CentraleSupélec

- *Coursework.* Machine Learning, Algorithms, Advanced Probability, Statistics, Stochastic Calculus, Harmonic Analysis.

Classes Préparatoires aux Grandes Écoles — Mathematics & Physics

2016 – 2018

Strasbourg, France

Lycée Kléber

- Two-year intensive program preparing the French national engineering entrance exams.

RESEARCH EXPERIENCE

Ph.D. Researcher

Jul 2023 – Present

Paris / Nice, France

Criteo AI Lab & CNRS / Université Côte d’Azur

- Research on 3D computer vision: generative modeling, 3D reconstruction, scene representation.
- First- or co-first-author on four papers (two at ICLR, one at TMLR, one under review).
- Co-supervised Master’s interns and open-sourced code accompanying each publication.

Research Intern — Generative 3D Computer Vision

2022 – 2023 (6 mo.)

Paris, France

Criteo AI Lab

- Generative 3D modeling with Neural Radiance Fields, adversarial networks, and diffusion models.
- Internship extended directly into a Ph.D. under the same team.

Research Intern — Computer Vision

2021 (6 mo.)

Paris, France

Schlumberger AI Lab

- Fine-tuned state-of-the-art ResNet-based architectures for industrial vision tasks.
- Experimental work on GAN-driven data augmentation for image segmentation networks.

Machine Learning Intern

2020 – 2021 (6 mo.)

Aarhus, Denmark

Aarhus University

- Integrated a GAN into a citizen-science creativity game; designed Bayesian models to infer player behavior.
- Optimized a quantum cold-atoms experiment using Gaussian Process regression.

PUBLICATIONS

* equal contribution

Peer-reviewed

1. SphericalDreamer: Generating Navigable Immersive 3D Worlds with Panorama Fusion

A. Schnepf, K. Kassab, F. Vasile, A. Comport

ICML 2026 · sphericaldreamer.github.io

2. Fused-Planes: Why Train a Thousand Tri-Planes When You Can Share?

K. Kassab*, A. Schnepf*, J.-Y. Franceschi, L. Caraffa, F. Vasile, J. Mary, A. Comport, V. Gouet-Brunet

ICLR 2026 · fused-planes.github.io

3. Bringing NeRFs to the Latent Space: Inverse Graphics Autoencoder

A. Schnepf*, K. Kassab*, J.-Y. Franceschi, L. Caraffa, F. Vasile, J. Mary, A. Comport, V. Gouet-Brunet

ICLR 2025 · ig-ae.github.io

4. RefinedFields: Radiance Fields Refinement for Planar Scene Representations

K. Kassab, A. Schnepf, J.-Y. Franceschi, L. Caraffa, J. Mary, V. Gouet-Brunet

TMLR 2025 · refinedfields.github.io

Preprint & Workshop

5. 3DGEN: A GAN-based Approach for Generating Novel 3D Models from Image Data

A. Schnepf, F. Vasile, U. Tanielian

NeurIPS Workshop on Machine Learning for Creativity and Design, 2022

TECHNICAL SKILLS

- *Research areas.* 3D computer vision, neural radiance fields (NeRFs), generative modeling, inverse graphics.
- *Mathematics.* Probability, statistics, stochastic calculus, harmonic analysis, optimization.
- *Programming.* Python, PyTorch, JAX, NumPy, Git, Linux.
- *Frameworks.* Nerfstudio, Hugging Face Diffusers, scikit-learn.

LANGUAGES

- French (native), English (fluent).